01 The life science tools framework

FRAMEWORK

A product framework spanning academic research, biotech, pharma, CRDMOs and manufacturing.

Reagents

01	Recombinant proteins / antibodies
02	Enzymes / PCR and NGS reagents
03	DNA / RNA / chemical building blocks
04	Cell lines / culture media
05	Assay reagents for R&D and QC

Tests × dose per test × price

Consumables

01	Tips / microplates
02	Culture flasks / centrifuge tubes
03	Membranes / functional beads
04	Columns / chromatography resins
05	Gels / process components

Tests or batches × intensity × price

Instruments

01	Sequencers / liquid handlers
02	LC / LC-MS / mass spectrometers
03	Flow cytometers
04	High-content imaging systems
05	Cell-culture workstations

New installations + replacement + service

Use one product taxonomy; retain the scope of each market estimate.

Classify products separately from their use cases. Chromatography resins can be reused.

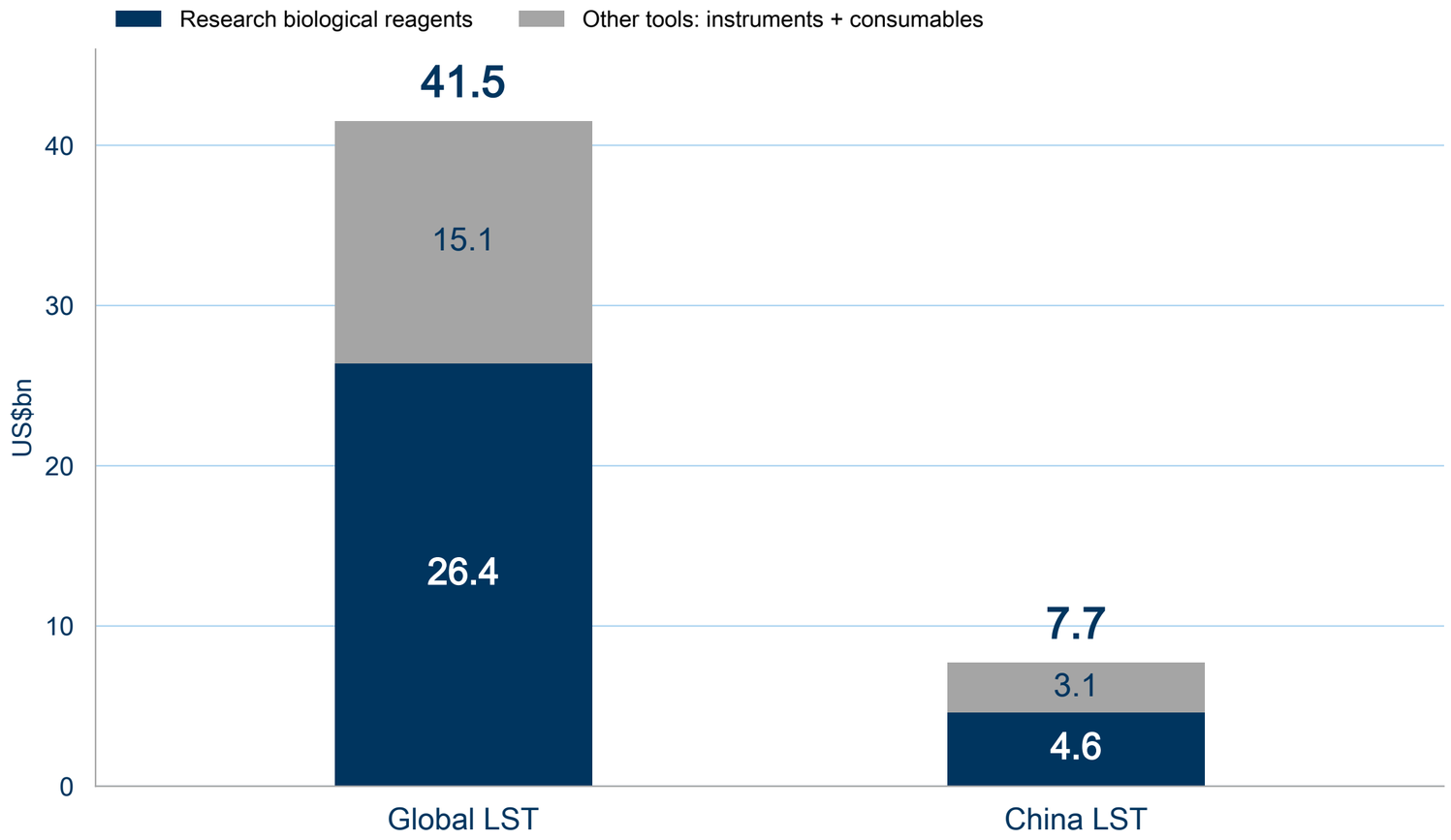
Source: supplied framework and prior product research. Existing TAMs do not automatically reconcile.

2026-10-12

02 China and global LST markets

HISTORICAL

2025 historical estimates from one F&S study; this is a defined market scope, not a universal LST TAM.



China / global market

18.6%

7.7 / 41.5
Same-scope historical estimate

Broader research and production markets require a separate scope reconciliation.

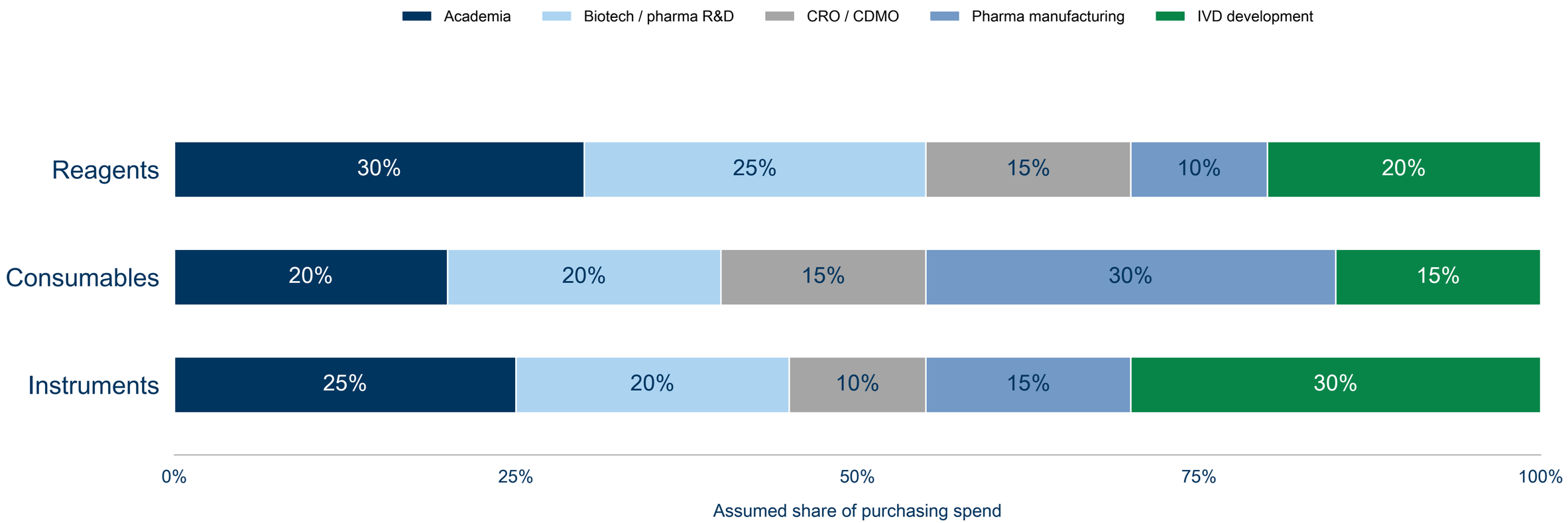
Other tools are the residual. Comparable instrument and consumable splits were not separately available.

Source: ACROBiosystems HKEX application, 30 Sep 2026, printed pp. 65 and 67; links in SOURCES.md.

03 How downstream demand differs

ILLUSTRATIVE

Direct purchasing differs by product basket. These supplied assumptions sum to 100% for each row.



Reagents: more research exposure in this scenario

Consumables: more production exposure

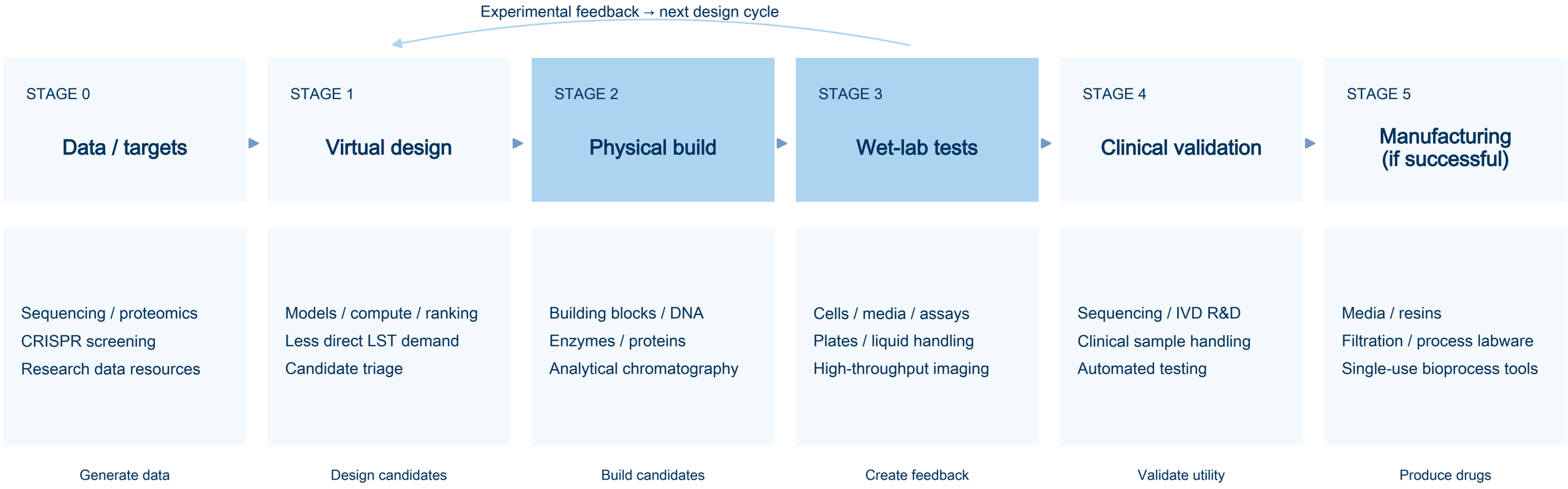
CRO/CDMO purchases are separate from pharma-owned production. Hospitals are not separately modeled.

Source: illustrative shares in the supplied manuscript; these are not measured customer shares.

04 AI discovery and its LST touchpoints

MECHANISM

AI candidates need physical construction and experimental feedback. Orders depend on funded projects.



The most direct potential uplift comes from building and testing candidates.

Stages 2–3 have more direct exposure. Clinical and manufacturing demand face attrition and time lags.

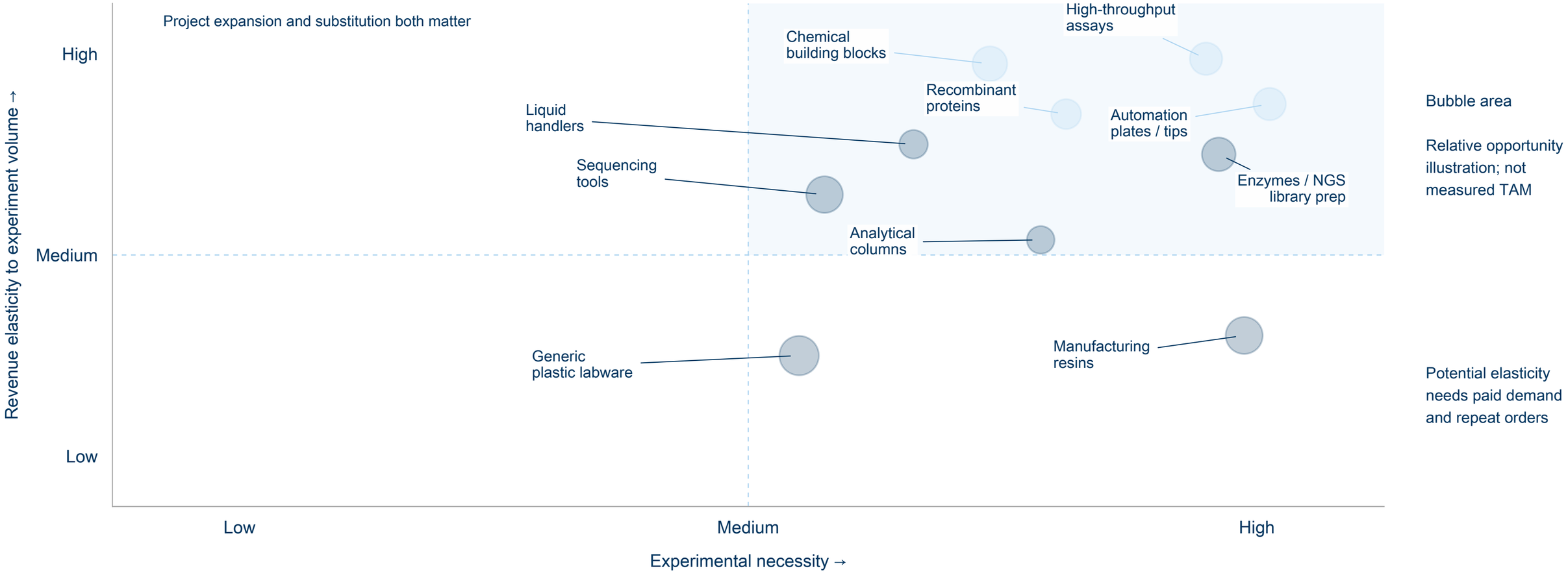
Source: supplied framework and product-use research. Arrows describe a mechanism, not realized orders.

2026-10-12

05 Which tools have more demand elasticity?

HYPOTHESIS

Positions and bubble areas are qualitative research judgments, not measured scores or additive TAMs.



Net demand also depends on consumption intensity, prices, project attrition and technical substitution.

Source: supplied framework and analyst assumptions. Edit positions and relative areas in data.json.

2026-10-12

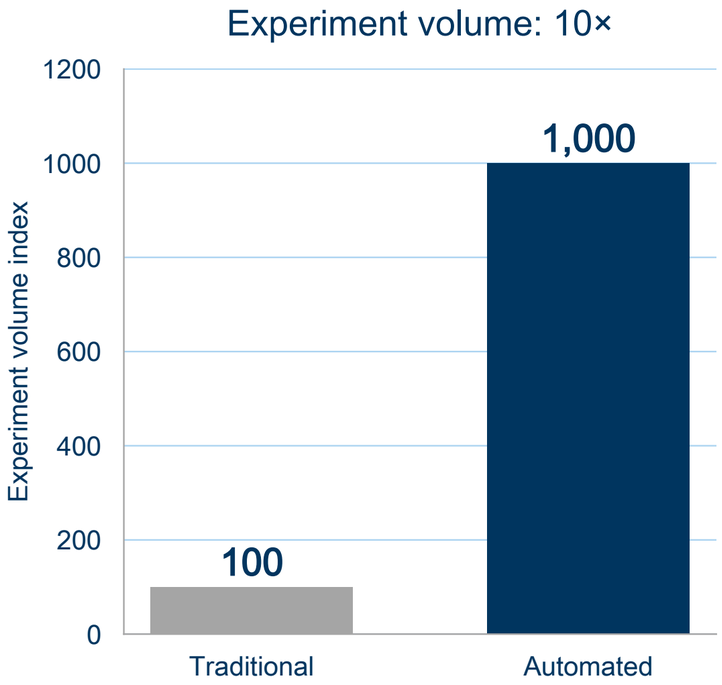
06 More experiments, less material per experiment

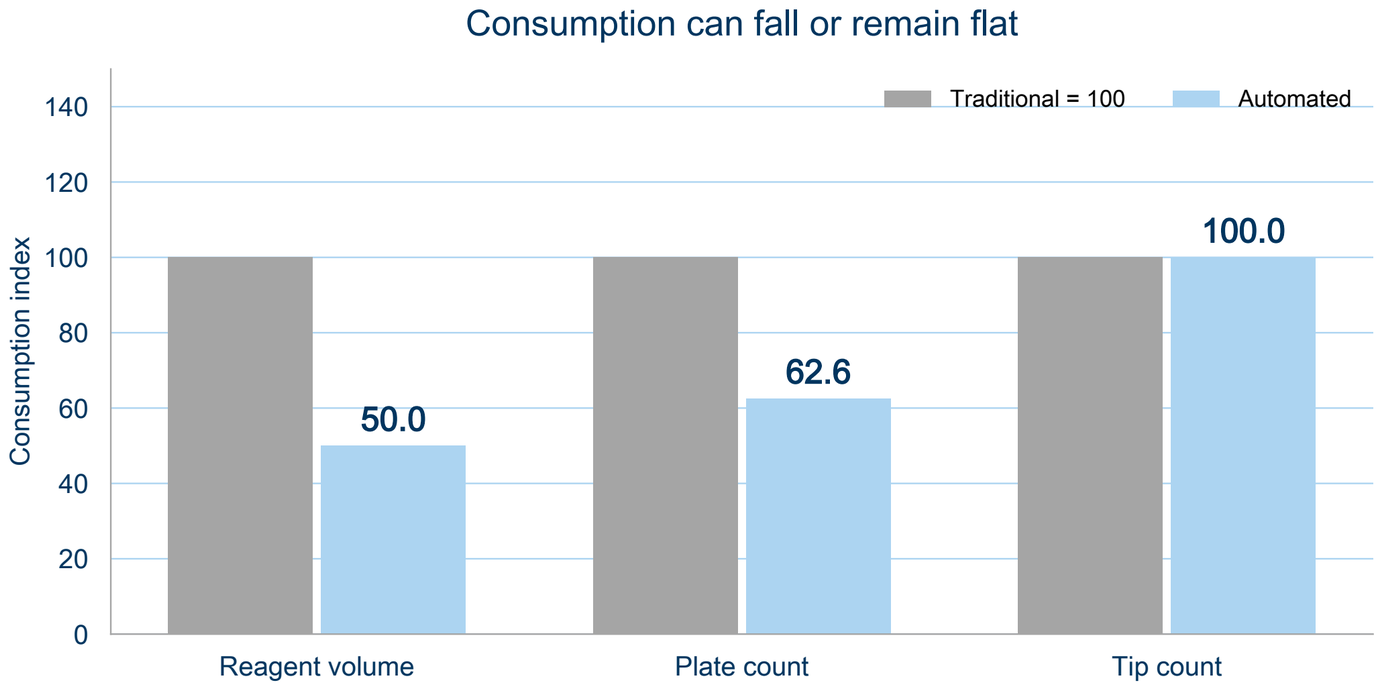
SCENARIO

Traditional: 100k tests, 96 wells, 100 µL/test. Automated: 1m tests, 1,536 wells, 5 µL/test. Base = 100.

100k → 1m experiments

Reagents: 10 L → 5 L · Plates: 1,042 → 652 · Tips: 100k → 100k (assumed)





Experiment growth does not imply proportional purchasing growth.

Y-axis scales differ. Full plates, no dead volume or control wells; tips fall from 1 to 0.1 per test. ASP excluded.

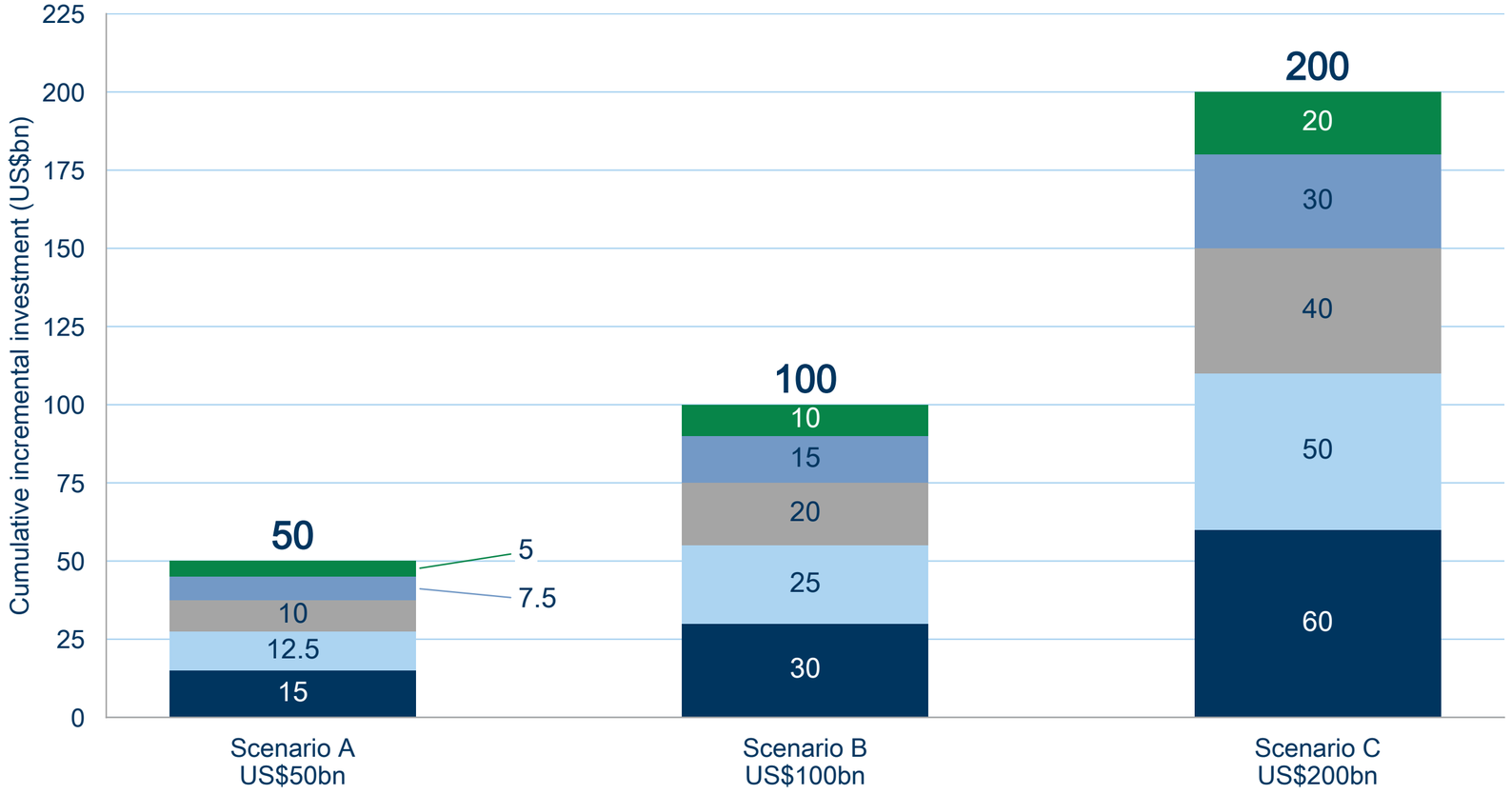
Source: supplied scenario and analyst calculation. Test definition and plate utilization affect the result.

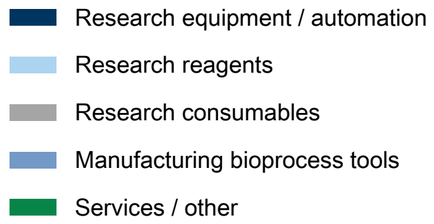
2026-10-12

07 Illustrative AI-driven investment scenarios

SCENARIO

US$50bn / 100bn / 200bn are illustrative cumulative investment pools; no time horizon is specified.





Build phase → installed equipment

Operate → recurring purchases

Manufacturing needs drug output.

Allocations: 30% / 25% / 20% / 15% / 10%. Research and manufacturing buckets are mutually exclusive.

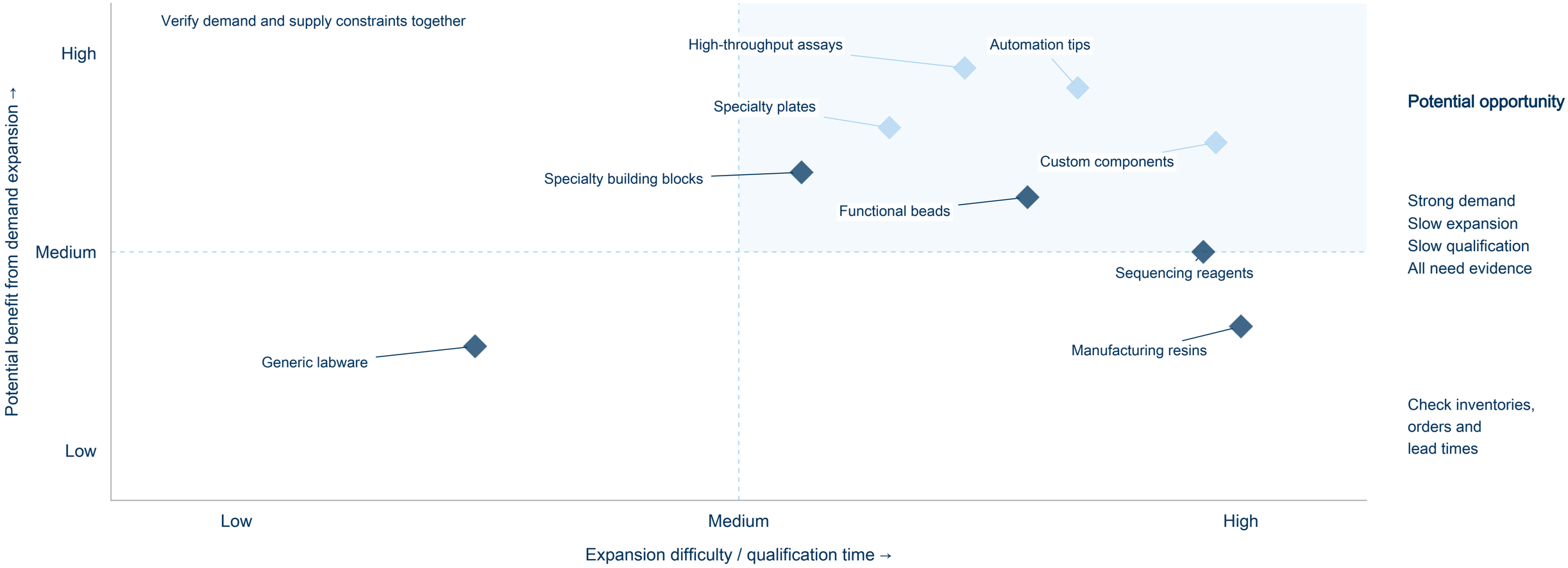
Source: supplied scenario and allocation assumptions. These are not realized funding, orders or annual TAMs.

2026-10-12

08 Potential supply constraints over 12–24 months

HYPOTHESIS

Relative expansion and qualification constraints over 12–24 months; current shortages are not established.



Track lead times, stockouts, qualification, yields and prices. Manufacturing resins have a longer demand lag.

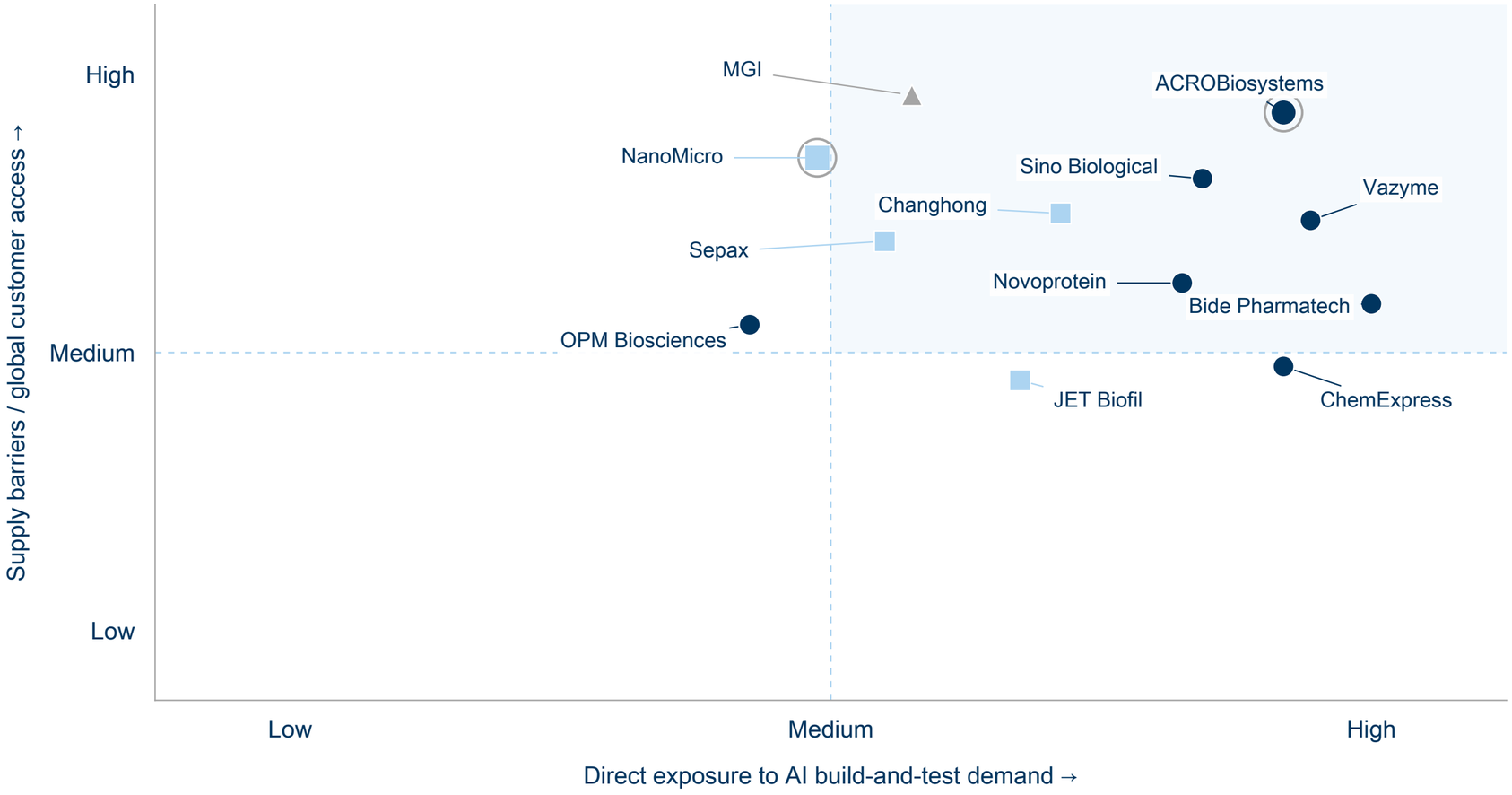
Source: supplied framework and analyst hypotheses. Technical barriers do not prove a capacity bottleneck.

2026-10-12

09 Mapping Chinese listed LST suppliers

HYPOTHESIS

Product exposure to AI build-and-test stages; positions do not measure AI revenue share or investment rank.



Reagents

Proteins, enzymes, blocks, media

Consumables

Labware, columns, resins

Instruments / platforms

Sequencing and automation

Two different demand pathways

ACRO: expression / functional assays
NanoMicro: purification + scale-up

Axes are qualitative judgments. Qualification, global customer access and paid orders still require evidence.

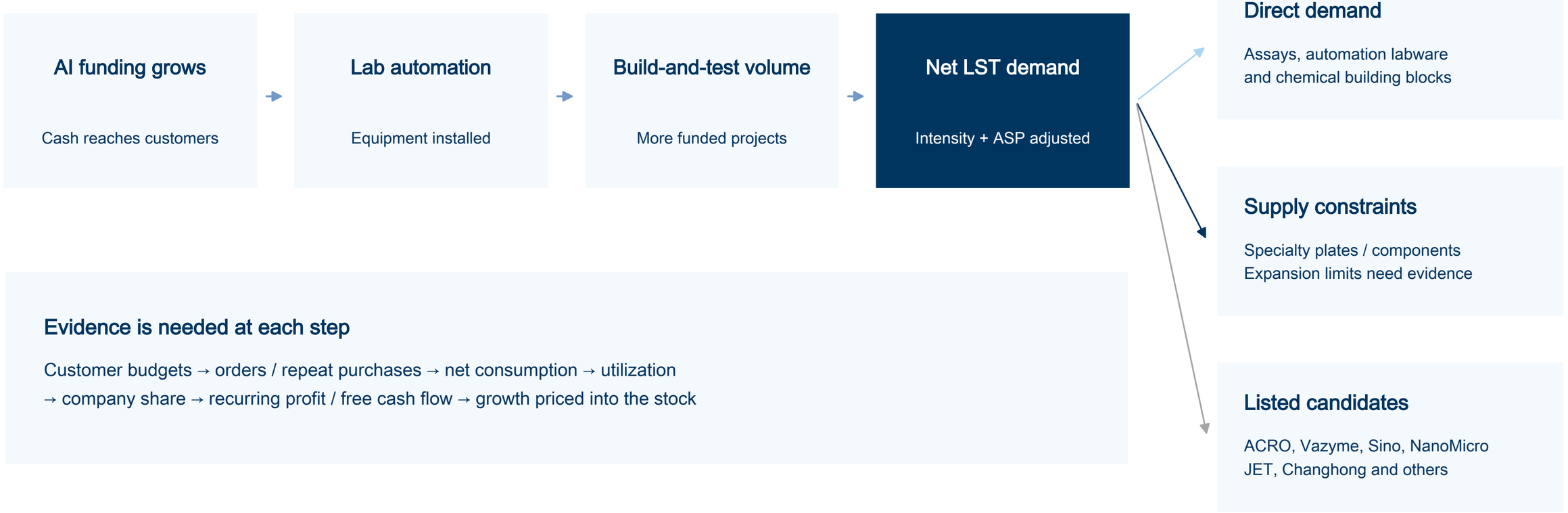
Sources: company filings and official product catalogs. Product rationale and assumptions are in data.json.

2026-10-12

10 From AI funding to LST shareholder returns

THESIS

Funding starts the chain. Projects, consumption, supply, market share and cash returns determine the outcome.



Evidence is needed at each step

Customer budgets → orders / repeat purchases → net consumption → utilization
→ company share → recurring profit / free cash flow → growth priced into the stock

Experimental relevance ∩ Evidenced supply constraints ∩ Global customer access

Seek experimental relevance, evidenced supply constraints and global access; test the valuation hurdle.

Source: supplied framework and prior company research. Capital announcements are not LST orders.

2026-10-12

Design capacity expands. Validation must keep up.

Illustrative stage comparison, normalized per one clinical entrant. Supplied assumptions, not cohort statistics.

STAGE	PRE-AI	AI ERA	WHAT CHANGES
Molecules designed (virtual)	~10^3	10^5–10^9	Generative models can expand candidate enumeration at low marginal cost.
Molecules physically made	~10^3	10^3–10^4	Automated synthesis and DNA libraries can expand construction capacity.
Fully profiled in wet lab	**~50**	**50 → 500**	**Illustrative capacity expansion that investors would need to fund.**
Clinical entry	1	1	Patients, ethics, capital and time remain hard constraints.
Expected approvals	~0.14	Unknown	Clinical success must be demonstrated; AI does not establish an approval rate.

Working thesis: experimental throughput becomes a capacity gate as virtual design scales.

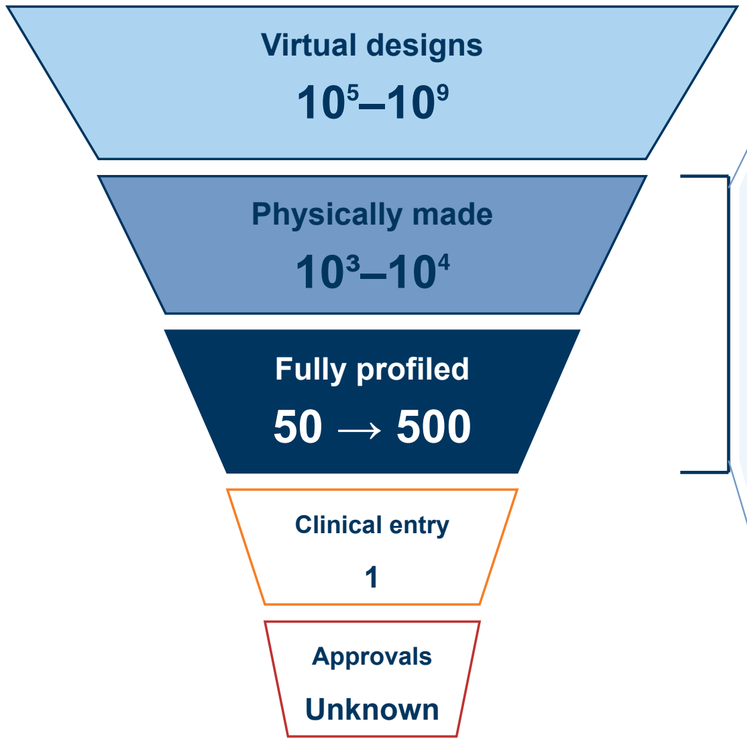
Source: user-supplied comparison. ~0.14 is an illustrative expected approval count per clinical entrant; the AI-era outcome is unknown.

AI-scale design meets finite experimental capacity

Working scenario: candidate generation expands faster than physical construction and biological validation.

AI-ERA DISCOVERY FUNNEL

Illustrative counts per one clinical entrant



WET LAB = BUILD + TEST

Funnel widths are schematic.
Clinical success remains unproven
in this scenario.

The LST ecosystem behind the capacity gate

Selected market anchors · different scopes and years · do not add these values

Molecular construction

DNA / RNA synthesis, enzymes
Chemical building blocks

RMB 1.33bn

China • 2025 • DNA synthesis services [1]
Oligos + long fragments only

TWIST BIOSCIENCE · GenScript Make Research Easy

CHINA / HK LISTINGS	US LISTINGS / PARENTS
GenScript · 1548.HK	Twist · TWST
Vazyme · 688105.SH	Thermo Fisher · TMO
Tsingke · NEEQ:874813	

Protein & functional assays

Recombinant proteins, antibodies
Binding and activity assays

US$3.5bn

Global • 2025 • research proteins [2]
China: ~US$0.4bn • RUO scope

Acro Biosystems

CHINA / HK LISTINGS	US LISTINGS / PARENTS
ACRO · 301080.SZ	Bio-Techne · TECH
Sino · 301047.SZ	Thermo Fisher · TMO

Cells & culture systems

Cell models, media, feeds
Bioreactors and cell expansion

US$323.5m*

China • 2024 • bioprocess media [3]
Selected media anchor; not all cell tools

Gibco / HyClone

CHINA / HK LISTINGS	US LISTINGS / PARENTS
OPM · 688293.SH	Gibco / Thermo · TMO
ACRO · 301080.SZ	Cytiva / HyClone · DHR

Labware & sample handling

Tips, plates, tubes, filtration
Automation-compatible labware

~RMB 23bn

China • 2024 • plastic labware [4]
Includes clinical and industrial uses

BIOFIL · BD

CHINA / HK LISTINGS	US LISTINGS / PARENTS
JET Biofil · 688026.SH	Corning · GLW
Changhong · 300151.SZ	BD · BDX

Measurement & analytics

Sequencing, LC / LC-MS, flow
Imaging and quality control

RMB 4.3bn

China • 2024 • sequencing tools [5]
Sequencers + consumables; all uses

MGI · BD flow → WAT†

CHINA / HK LISTINGS	US LISTINGS / PARENTS
MGI · 688114.SH	Illumina · ILMN
NanoMicro · 688690.SH	Agilent · A
	Waters · WAT†

Purification & scale-up

Analytical columns, process resins
Filtration and single-use systems

US$578.2m*

China • 2024 • bioprocess resins [3]
Manufacturing demand has a longer lag

NanoMicro · cytiva

CHINA / HK LISTINGS	US LISTINGS / PARENTS
NanoMicro · 688690.SH	Cytiva · DHR
Sepax · 688758.SH	Repligen · RGEN

Automation + data orchestration span every tool family — TECAN. SIX: TECN

† Ex-BD biosciences / flow moved to Waters on 9 Feb 2026; BD (BDX) here means specimen collection.

Groups denote listings, not incorporation; brands use parent tickers. * Source access limited. Sources and ticker register: SOURCES.md.

Fewer molecules. Deeper experimental data.

AI may reduce candidate counts while increasing characterization per molecule

From molecule throughput to data density: a potential driver of life science tools demand

Existing data: staged screening

1,000 × 1 initial label

Illustrative early single-task dataset

Antibody	Binds?
Ab-001	Yes
Ab-002	No
Ab-003	Yes
Ab-004	Yes
Ab-005	No
Ab-006	Yes

Binding first; deeper tests for selected hits

Multi-assay characterization already exists

AI: richer, linked training data

300–1,000 antibodies · Up to 9 label types

Measure selected properties on shared molecules; fill the information gaps

Antibody	KD	kon	koff	expr	Tm	agg	poly	epi	cell
Ab-001	○	●	●	●	●	●	●	●	●
Ab-002	○	●	●	●	●	●	●	●	●
Ab-003	○	●	●	●	●	●	●	●	●
Ab-004	○	●	●	●	●	●	●	●	●
Ab-005	○	●	●	●	●	●	—	●	—
Ab-006	○	●	●	●	●	●	●	—	●

Learn binding, expression, stability and functional relationships

● Measured label ○ KD derived from shared kinetics — Unmeasured; colors show coverage

Reuse existing data

Normalize formats, conditions and access
Better training may need no new experiment

Stage 0: fill data gaps

Add missing assays and biological diversity
Create reusable model-training datasets

Active learning: measure selectively

Prioritize informative molecules and tests
Limit coverage of costly measurements

Wet-lab demand depends on funded experiments, assay depth and net tool consumption

Illustrative counts. Nine label types are not nine independent assays; richer data need not come at unchanged cost.

Sources: Jain et al., 2017; Arsiwala et al., 2025; van Niekerk et al., 2026. Mechanism illustration; definitions and links in SOURCES.md.

Experimental data density and the LST ecosystem

Stage 0: biological data production creates a wet-lab use case

Learning developability requires comparable assay data and an integrated tool workflow

Stage 0: biological data production

Continuous input to training and iteration

Binding kinetics
KD / kon / koff

Expression & physicochemical properties
expression / Tm / aggregation

Specificity & cell function
polyreactivity / epitope / cell potency

Training records: sequence, target, conditions, raw curves, replicates, batches and failures

Train / improve → **Design / validate**

Assays inform models

Labels → assays → purchases

Tool revenue needs a bridge from label coverage to workflow volume and unit consumption

Construction & expression

DNA synthesis, vectors, expression, purification

Expression labels + samples for other assays

Count shared sample preparation once

Protein & functional testing

SPR / BLI, binding curves, epitope binning

Target proteins, sensors / chips, buffers

KD / kon / koff can share one kinetic run

Cells & culture systems

Functional cells, media, transfection, expansion

Cell potency; selected polyreactivity assays

Record cell model, target level and conditions

Measurement & analysis

SEC / HIC, nanoDSF, LC-MS, flow cytometry

Stability, aggregation, characterization / QC

Columns are reused; model instrument CAPEX separately

Sample handling, labware & automation

Plates, tips, tubes, filtration / beads, liquid handling

Across all workflows; miniaturization, reuse and scheduling alter net consumption

ACROBiosystems: target proteins, functional cells & characterization

AIDD-Ready offering exists; paid orders and repeat purchases still need verification

NanoMicro: analytical separation & lab purification materials

Track sample runs, column life and ASP; manufacturing resin demand has a longer lag

LST purchases ≈ Σ [workflow executions × net purchased inputs per run × ASP]

Instrument CAPEX is separate. Product alignment does not establish AI orders or revenue. Sources: Ginkgo, Sartorius and company websites.

Typical protein / antibody screening workflow

From DNA constructs to functional evidence: the life science tools behind each step

Example: sequence-defined antibody candidates screened against a protein target

BUILD · Make physical samples | **TEST · Find binders** | **AI learning needs depth per molecule** | **LEARN · Iterate**

Potential constraint: comparable, reproducible assay data

01 / BUILD

Construct design / DNA synthesis

Encode antibody candidates as DNA constructs

OUTPUT
Sequence + construct

REAGENTS
Oligos / gene fragments
PCR enzymes, vectors
Cloning reagents

INSTRUMENTS
Thermocycler
Spectrophotometer
Liquid handling workstation

CONSUMABLES
PCR / deep-well plates
Tubes, pipette tips

REPRESENTATIVE SUPPLIERS
TWIST BIOSCIENCE · GenScript Make Research Easy
Tsingke · Vazyme
Thermo Fisher

02 / BUILD

Protein expression & purification

Produce candidate antibodies and target-protein samples

OUTPUT
Purified samples + yield

REAGENTS
Cells, media, transfection
Affinity resin, buffers

INSTRUMENTS
Shaker / bioreactor*
ÄKTA / FPLC system
Centrifuge

CONSUMABLES
Flasks / culture bags*
Columns, filters, tubes

REPRESENTATIVE SUPPLIERS
cytiva
GenScript · ACROBiosystems
Sino Biological · Gibco
OPM · Duoning

03 / TEST

Primary binding screen

Identify binders and rank initial binding signals

OUTPUT
Hit / binding signal

REAGENTS
Recombinant target proteins
Antibody candidates
Detection reagents, buffers

INSTRUMENTS
ELISA / plate reader
BLI or SPR platform

CONSUMABLES
96 / 384-well plates
BLI sensors / SPR chips
Sealing films, sample tubes

REPRESENTATIVE SUPPLIERS
ACRO BIOSYSTEMS
Sino Biological · Bioss
Sartorius · Cytiva · Tecan
JET Biofil

04 / TEST

Multi-property characterization

Evaluate binding, specificity and developability risks

KD · kon · koff · expression†
Tm · aggregation
polyreactivity · epitope binning

REAGENTS
Reference proteins, dyes
Assay panels, buffers
Analytical reagents

INSTRUMENTS
SPR / BLI; SEC-HPLC
LC / LC-MS
nanoDSF / DLS

CONSUMABLES
Columns, capillaries
Vials, filters, assay plates
Sensors / chips

REPRESENTATIVE SUPPLIERS
NanoMicro
Waters · Agilent · Thermo Fisher
NanoTemper · Sepax
SPR / BLI: Cytiva · Sartorius

05 / TEST

Cell-function validation

Test activity in a relevant cellular model

LABEL
Cell potency

REAGENTS
Cell models, media, cytokines
Staining antibodies
Reporter-assay reagents

INSTRUMENTS
Flow cytometer
High-content imaging
Plate reader

CONSUMABLES
Cell-culture plates / flasks
Pipette tips, filters

REPRESENTATIVE SUPPLIERS
BIOFIL®
Gibco · OPM · Vazyme · Bio-Techne
Molecular Devices · Corning
Waters Biosciences (ex-BD)

06 / LEARN

Data integration & feedback

Link assay results to the next AI design

OUTPUT
Traceable training data

REAGENTS
No direct assay reagents

SYSTEMS / SOFTWARE
LIMS / ELN
Linked data platform
Lab orchestration

CONSUMABLES
No direct assay consumables

REPRESENTATIVE SUPPLIERS
TECAN.
Hamilton
Benchling · Genedata

Sequence + assay context + results → next design cycle

Automation + data orchestration span every step

Wet lab = build + test. AI may test fewer molecules while increasing characterization depth per molecule.

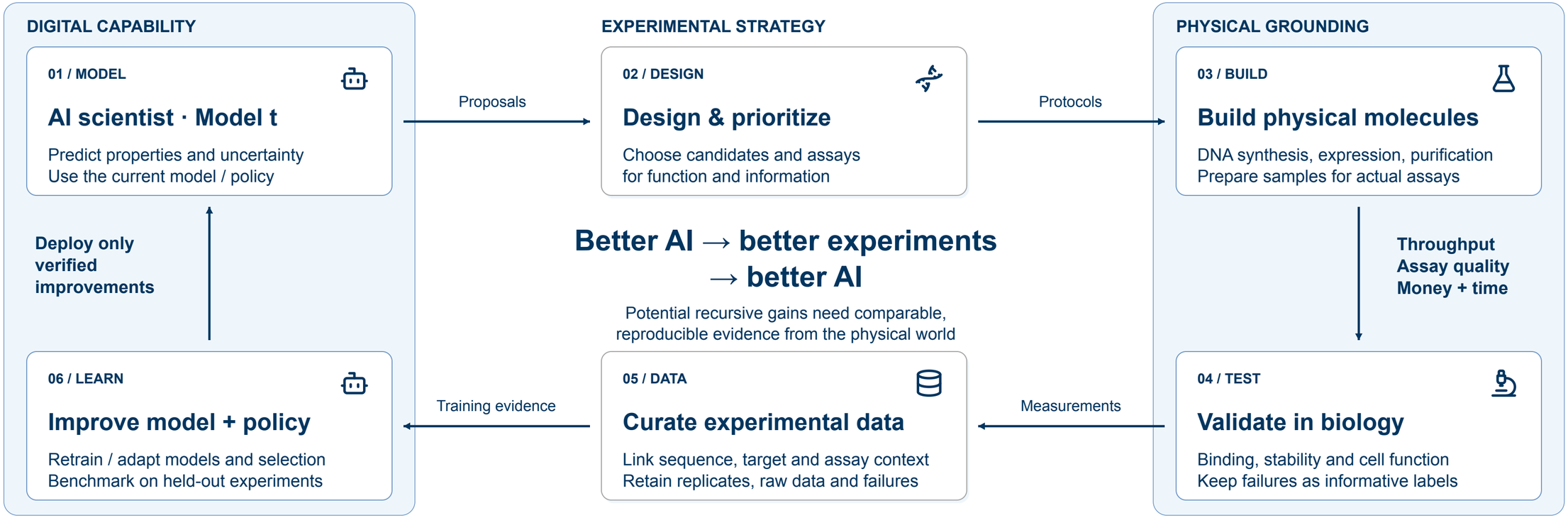
LST demand depends on molecule count × assay depth × net inputs per assay; instrument investment is separate.

Illustrative workflow; assays may run in parallel. * Scale-dependent. † Expression comes from stage 2. Supplier examples and primary sources: SOURCES.md.

RSI: AI design and biological validation

How experimental feedback could improve both biological designs and the AI design process

RECURSIVE SELF-IMPROVEMENT · CONCEPTUAL MECHANISM



LST supports every return to physical biology

BUILD
DNA synthesis · expression · purification

TEST
Proteins · biosensors · analytical columns
Cell assays · measurement platforms

CONNECT + SCALE
Labware · automation · data platforms
Comparable records across batches

The pace of improvement depends on useful experimental evidence per cycle

Potential LST demand ∝ funded cycles × assay depth × net inputs per assay

Automation and miniaturization can reduce inputs per measurement; instrument CAPEX is separate.

Conceptual RSI mechanism. Closed-loop learning does not establish sustained autonomous RSI. Primary sources: SOURCES.md.